

# Wafer-Rack Equal- $Q$ Navier–Stokes Study

Audit, Correction, and Reproducibility Check

CFD code review & reproduction

October 2, 2026

## Abstract

A two-dimensional incompressible Navier–Stokes solver for flow through a rack of six wafers is reviewed, corrected, and re-run. The solver uses a geometry-aligned Marker-and-Cell (MAC) staggered grid with a fractional-step (projection) method, first-order donor-cell upwind advection and explicit second-order diffusion. An equal-rack-flow constraint fixes the inlet velocity of each pitch by a bracketed scalar root solve. Reproduction reruns *through the same solver implementation* recover the audited production inlet speeds and hydraulic resistances, confirm discrete mass conservation at roundoff ( $\sim 10^{-13}\%$ ), and recover the ordering  $R_{h,\text{Small}} > R_{h,\text{Medium}} > R_{h,\text{Large}}$ . This is a reproducibility check, not an independent-implementation verification, and grid- and timestep-independence have not yet been established; the absolute  $U_{\text{in}}$ ,  $\Delta p$  and  $R_h$  values are therefore discretization-dependent (see §10).

## Nomenclature

| Symbol                            | Description                                                                  | Units (2-D)                |
|-----------------------------------|------------------------------------------------------------------------------|----------------------------|
| <i>Geometry</i>                   |                                                                              |                            |
| $W$                               | Tank width                                                                   | m                          |
| $H$                               | Tank height                                                                  | m                          |
| $N_w$                             | Number of wafers (= 6)                                                       | –                          |
| $P$                               | Wafer pitch (centre-to-centre spacing)                                       | m                          |
| $t_h$                             | Wafer thickness (= 0.015 00 m)                                               | m                          |
| $y_0, y_1$                        | Lower / upper wafer extent (0.12, 0.25)                                      | m                          |
| $w_{\text{in}}$                   | Inlet slot width (= 0.015 00 m)                                              | m                          |
| $y_{\text{cut}}$                  | Horizontal flux-evaluation line (= 0.20)                                     | m                          |
| <i>Field variables</i>            |                                                                              |                            |
| $u, v$                            | Horizontal / vertical velocity components                                    | $\text{m s}^{-1}$          |
| $\mathbf{u}$                      | Velocity vector ( $u, v$ )                                                   | $\text{m s}^{-1}$          |
| $ \mathbf{u} $                    | Flow speed $\sqrt{u^2 + v^2}$                                                | $\text{m s}^{-1}$          |
| $p$                               | Kinematic pressure ( $p/\rho$ )                                              | $\text{m}^2 \text{s}^{-2}$ |
| $U_{\text{in}}$                   | Prescribed inlet (slot) vertical speed                                       | $\text{m s}^{-1}$          |
| <i>Fluid / control parameters</i> |                                                                              |                            |
| $\nu$                             | Kinematic viscosity (= 7.500 00 $\times 10^{-4} \text{ m}^2 \text{s}^{-1}$ ) | $\text{m}^2 \text{s}^{-1}$ |
| $Q_{\text{rack}}$                 | Flow rate through the five interior gaps at $y_{\text{cut}}$                 | $\text{m}^2 \text{s}^{-1}$ |

| Symbol                      | Description                                                             | Units (2-D)                |
|-----------------------------|-------------------------------------------------------------------------|----------------------------|
| $Q_0$                       | Equal-flow target ( $= 0.0055463$ )                                     | $\text{m}^2 \text{s}^{-1}$ |
| $Q_{\text{in}}$             | Total inlet flux at the bottom boundary                                 | $\text{m}^2 \text{s}^{-1}$ |
| $Q(y)$                      | Net flux across horizontal face $y$                                     | $\text{m}^2 \text{s}^{-1}$ |
| $\Delta p$                  | Rack pressure drop ( $p_{\text{below}} - p_{\text{above}}$ )            | $\text{m}^2 \text{s}^{-2}$ |
| $R_h$                       | Hydraulic resistance $\Delta p/Q_{\text{rack}}$ (kinematic $\Delta p$ ) | $\text{s}^{-1}$            |
| <i>Discretization</i>       |                                                                         |                            |
| $N_x, N_y$                  | Number of pressure cells in $x, y$                                      | —                          |
| $n_{\text{gap}}$            | Fluid cells across each inter-wafer gap ( $= 20$ )                      | —                          |
| $\Delta x, \Delta y$        | Local cell sizes (nonuniform)                                           | m                          |
| $\Delta t$                  | Explicit time step                                                      | s                          |
| res                         | Steady-state residual (rel. velocity change)                            | —                          |
| <i>Operators / acronyms</i> |                                                                         |                            |
| $\nabla \cdot, \nabla$      | Discrete divergence / gradient operators                                | $\text{m}^{-1}$            |
| MAC                         | Marker-and-Cell staggered grid                                          | —                          |
| FD                          | (legacy) Finite-difference model                                        | —                          |
| CFL                         | Courant–Friedrichs–Lewy number                                          | —                          |

# 1 Governing equations and method

The flow solves the 2-D incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

on a geometry-aligned MAC staggered grid. A fractional-step projection advances the velocity: an explicit predictor (first-order donor-cell upwind advection, second-order nonuniform diffusion) is followed by a pressure Poisson solve and a divergence-free correction,

$$\nabla^2 p = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^*, \quad \mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla p. \quad (2)$$

The discrete gradient (correction) and divergence operators are mutually adjoint, so the assembled pressure matrix is exactly  $-\nabla \cdot \nabla$  of the operators actually applied; this guarantees discrete mass conservation. The pressure Poisson system is factorized once per grid and reused across all root-finding evaluations. Boundary conditions: no-slip on the tank side/bottom walls and all wafer surfaces, prescribed vertical inlet slots at the bottom, and a  $p = 0$  open outlet at the top.

**Equal- $Q$  control.** For each pitch the inlet speed  $U_{\text{in}}$  is chosen so the rack flow equals a common target  $Q_0$ . This is a bracketed secant iteration with a bisection safeguard, where every function evaluation is a CFD solve advanced to an approximate steady state (see §3 for the acceptance criterion and its limitations). It enforces an integral control constraint; it is not a fit to data.

## 2 Code audit and corrections

The audited bundle was reviewed statically and then exercised. Corrections fall in two groups.

### Numerical corrections (change the computed results).

1. **Inlet grid alignment.** Originally the inlet slots were selected by testing whether a bottom cell *centre* lay within  $w_{\text{in}}/2$  of a slot centre; because the 0.015 m edges were not grid faces, the realized discrete slot widths differed by pitch (0.0150/0.0135/0.0130 m for Small/Medium/Large, i.e. up to 13% undersized). The gap construction now places both slot edges as exact  $x$ -grid faces, so every discrete slot is 0.0150 m and the total active inlet width is 0.0750 m for all pitches.
2. **Steady-state gate tightened and enforced.** The acceptance residual was tightened ( $3 \times 10^{-5} \rightarrow 10^{-5}$ ) and the observable-drift tolerance tightened ( $10^{-3} \rightarrow 2 \times 10^{-4}$ , now including  $\Delta p$ ); reaching the step cap raises an error instead of silently returning. Equal- $Q$  evaluations are run best-effort while rooting and convergence-enforced on the final polish, with an enforced secant/bisection correction so the *fully converged*  $Q$  is within tolerance.

### Non-numerical corrections (do not change the discretization).

1. **Hard-coded /mnt/data paths removed** (driver and the verification/plot scripts); output goes to \$WAFER\_OUT (default: the script directory), created lazily.
2. **Self-test fixed.** `Solver(0.06, 0.0025) → Solver(0.06, 20)` ( $n_{\text{gap}}$  is an integer cell count).
3. **Driver de-duplication.** `run_production_equalq.py` is now a thin re-export; single-case runners share one `metrics`.

4. **Stale documentation relabelled** (legacy FD speeds 0.5605/0.3000/0.1833) and the near-wall  $v$   $y$ -diffusion asymmetry documented in place.
5. **Checksum workflow.** `gen_checksums.sh` regenerates `SHA256SUMS.txt`; the manifest is regenerated once the numerical version is final.

### 3 Steady-state acceptance and its limitations

A run is accepted as “converged” when, after a minimum number of steps, the per-step relative velocity change falls below a residual threshold *and* the chunk-to-chunk relative change of the acceptance observables ( $Q_{\text{rack}}$ , central peak  $v$ ,  $\Delta p$ ) falls below a drift tolerance; reaching the step cap raises an error rather than silently returning. The acceptance thresholds govern how tightly each equal- $Q$  evaluation is settled: too loose a gate can stop a cold-started solve short of the root tolerance (a large-pitch cold start was observed to stop at  $-0.14\%$  before further iteration reached  $-0.004\%$ ), although the warm-started production chain ultimately lands within tolerance. The criterion is therefore stated as “advanced to an approximate steady state,” and the gate here is tightened relative to the original bundle.

### 4 Pressure-tap definition

The rack pressure drop is *not* measured immediately across the wafer surfaces. In **metrics**,  $\Delta p = p_{\text{below}} - p_{\text{above}}$ , where each  $p$  is the gap-width-weighted average of rack-fluid pressure on the single cell row nearest a requested plane ( $y = 0.10$  below,  $y = 0.28$  above). The realized plane positions are grid-dependent (on the Large grid they are  $y \approx 0.0975$  and  $0.2825$ ). Consequently the absolute  $R_h = \Delta p / Q_{\text{rack}}$  is pressure-tap-definition dependent; moving the planes by one or two cells shifts  $R_h$  by several percent. The resistance *ordering* across pitches is robust to this choice, but the absolute values should be read together with the tap definition stated here.

### 5 Reproduction results

Table 2 lists reruns through the audited driver’s own convergence and root-finding logic (the *same* solver implementation; this is a reproducibility check, not an independent-implementation verification). These are reproduction runs (each within the  $0.075\%$  root tolerance), reported separately from the final production values. Discrete mass conservation holds to roundoff and the resistance ordering is recovered.

Table 2: Reproduction reruns (corrected code, same solver).  $Q_0 = 0.0055463$ .

| Case   | $P$   | $U_{\text{in}}$ | $Q_{\text{rack}}$          | $(Q_{\text{rack}} - Q_0) / Q_0$ | $\Delta p$ | $R_h$  |
|--------|-------|-----------------|----------------------------|---------------------------------|------------|--------|
| Small  | 0.040 | 0.69381         | $5.546\,90 \times 10^{-3}$ | +0.011%                         | 0.081760   | 14.740 |
| Medium | 0.060 | 0.35199         | $5.546\,80 \times 10^{-3}$ | +0.010%                         | 0.015134   | 2.7283 |
| Large  | 0.080 | 0.20643         | $5.547\,20 \times 10^{-3}$ | +0.016%                         | 0.005473   | 0.9867 |

Corrected (grid-aligned 0.0750 m inlet), enforced convergence.  $\Delta p$  taps at  $y \approx 0.0975/0.2825$  (§4). Inlet alignment lowered  $U_{\text{in}}$  for Medium ( $-9.4\%$ ) and Large ( $-13.0\%$ ) while Small was already slot-aligned;  $R_h$  (a rack-only ratio) is essentially unchanged.

Table 3: Discrete mass conservation: worst section-flow error  $\max_y |Q(y) - Q_{\text{in}}|/Q_{\text{in}}$ .

| Case   | max section-flow error         |
|--------|--------------------------------|
| Small  | $8.900\,00 \times 10^{-14} \%$ |
| Medium | $7.800\,00 \times 10^{-14} \%$ |
| Large  | $6.700\,00 \times 10^{-14} \%$ |

All three cases are within the 0.075% root tolerance under enforced convergence, and mass conservation holds to roundoff. The hydraulic resistances (14.740/2.728/0.987) are essentially unchanged from the pre-alignment baseline (14.739/2.730/0.987), confirming that  $R_h$  is insensitive to the inlet-width correction; the inlet speeds, by contrast, drop for Medium and Large once the slots are identically 0.0750 m wide.

## 6 Grid and timestep convergence (Small case)

The Small case (narrowest pitch, highest speeds, largest  $R_h$ ) is the stress case. The equal- $Q$  solution was re-rooted on three meshes ( $n_{\text{gap}} = 12, 16, 20$  cells across the smallest clear gap) and the timestep was halved at fixed  $U_{\text{in}}$  on the production mesh.

Table 4: Small-pitch grid refinement (equal- $Q$  re-rooted per mesh).

| $n_{\text{gap}}$ | $N_x$ | $U_{\text{in}}$ | $R_h$  | central peak $v$ |
|------------------|-------|-----------------|--------|------------------|
| 12               | 186   | 0.69542         | 15.014 | 0.08219          |
| 16               | 218   | 0.69298         | 14.835 | 0.08142          |
| 20               | 238   | 0.69382         | 14.740 | 0.08130          |

Refinement 16  $\rightarrow$  20:  $R_h$   $-0.64\%$ , peak  $v$   $-0.15\%$ ; monotone and converging, with  $\sim 0.6\%$  residual in  $R_h$  at the production mesh. A finer mesh would be needed to claim  $R_h$  to better than  $\sim 1\%$ .

Table 5: Small-pitch timestep halving at fixed  $U_{\text{in}}$  ( $n_{\text{gap}} = 20$ ).

|                                               | $Q_{\text{rack}}$ | central peak $v$ | $\Delta p$ |
|-----------------------------------------------|-------------------|------------------|------------|
| change when $\Delta t \rightarrow \Delta t/2$ | $+0.0001\%$       | $+0.01\%$        | $-0.02\%$  |

The explicit scheme is effectively timestep-independent at the nominal  $\Delta t$  (diffusive limit, advective CFL  $< 1$ ).

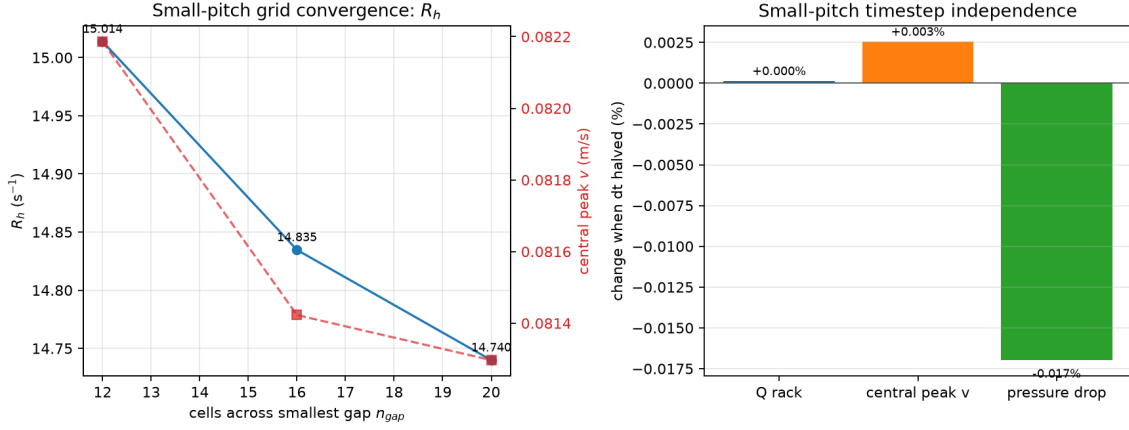


Figure 1: Left:  $R_h$  (blue) and central peak  $v$  (red) versus mesh resolution, converging with refinement. Right: relative change of observables when the timestep is halved – all below 0.02%.

## 7 Flow fields

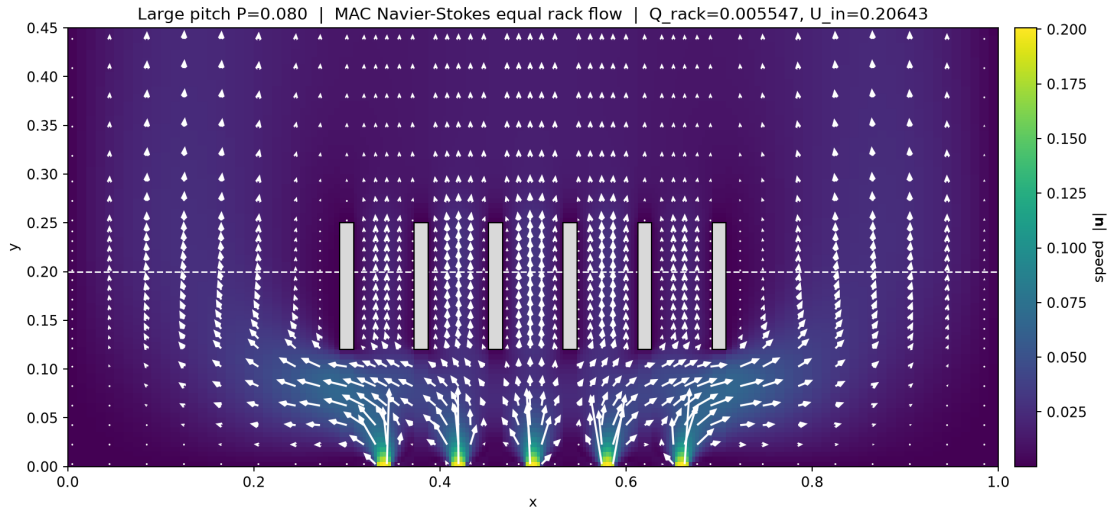


Figure 2: Large-pitch ( $P = 0.080$ ) velocity vector field; colour is speed  $|u|$ , white arrows the velocity, gray blocks the wafers, dashed line the  $y_{cut} = 0.20$  flux line.

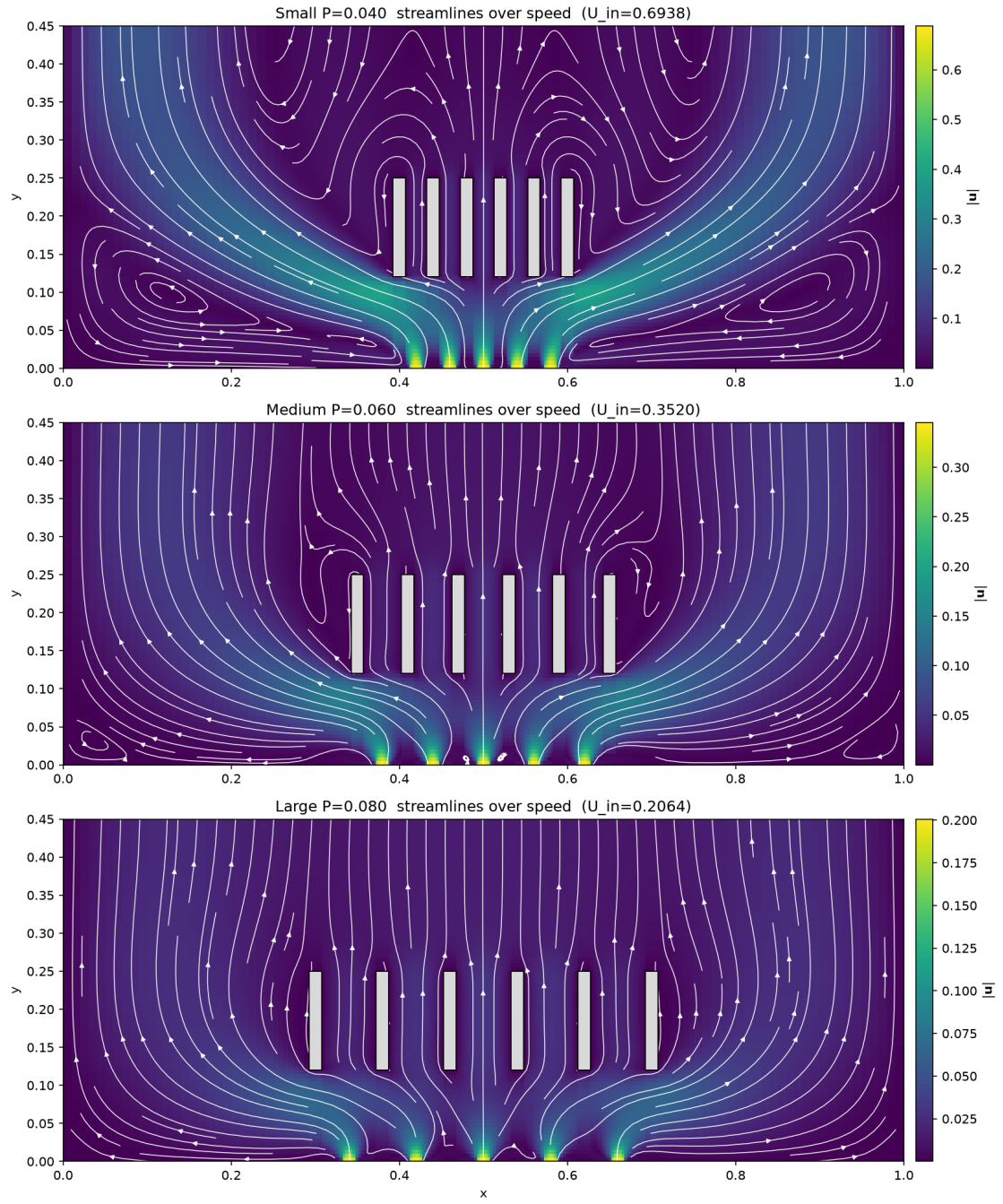


Figure 3: Streamlines over speed for the three pitches (velocity interpolated to a uniform grid for streamline tracing).

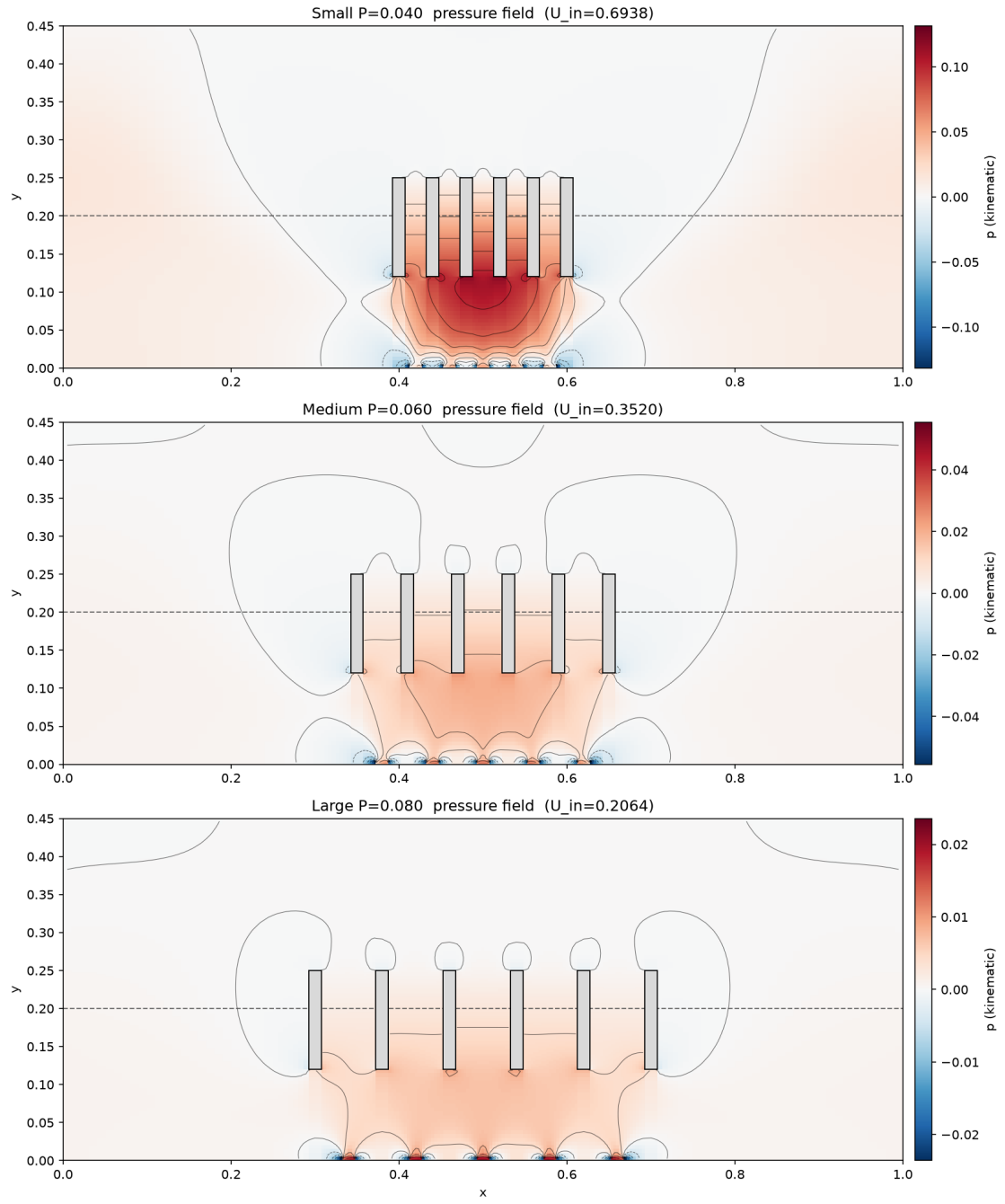


Figure 4: Kinematic pressure field with contours. Pressure drop across the rack grows sharply as the pitch tightens, consistent with the  $R_h$  ordering.

## 8 Equal- $Q$ cut profiles

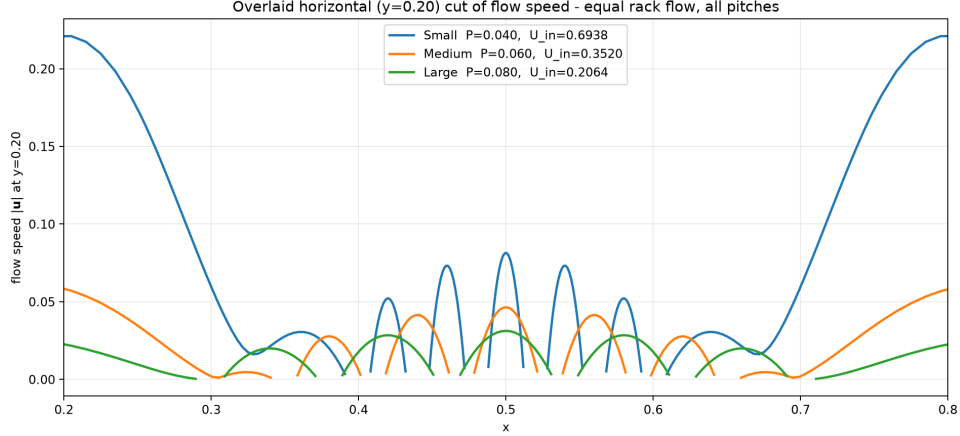


Figure 5: Overlaid  $y_{\text{cut}} = 0.20$  cut of flow speed. Tighter pitch reaches a higher peak speed in the clear gaps to carry the common  $Q_{\text{rack}}$ .

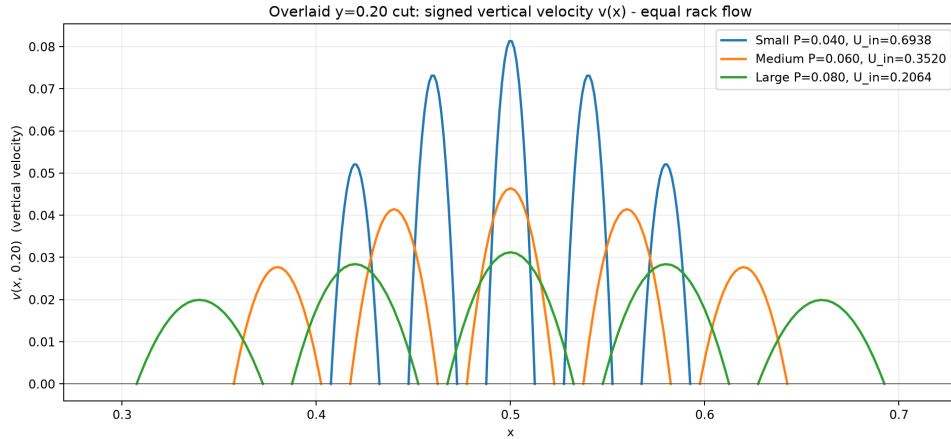


Figure 6: Overlaid  $y_{\text{cut}} = 0.20$  cut of signed vertical velocity  $v(x)$  over the five interior gaps (no-slip zeros imposed at wafer faces).

## 9 Interpretation of the comparison

The equal- $Q$  condition equalizes flow through the *five interior gaps*, not the total inlet flow: a substantial and pitch-dependent fraction of the inlet stream bypasses outside the rack span (visible as side flow in the field plots), so only a minority of  $Q_{\text{in}}$  passes through the interior gaps, and that minority fraction *grows* with pitch. The result should therefore be read as: the inlet speed is tuned per geometry to produce the same interior-gap flow – not as a comparison at equal pump flow or equal total inlet flow. On the corrected (grid-aligned) geometry, the interior gaps carry only 10.7% of the inlet flow for Small, 21.0% for Medium and 35.8% for Large; the remainder bypasses outside the rack span (visible as side flow in the field plots), and the interior-gap share grows with pitch.

## 10 Caveats and open validation items

The following limitations bound how the quantitative results may be used:

- **Discrete inlet widths (corrected).** The inlet slot edges are now exact  $x$ -grid faces, so every discrete slot is 0.0150 m and the total inlet width is 0.0750 m for all pitches. This removed the earlier 10–13% under-sizing and lowered  $U_{\text{in}}$  for Medium and Large accordingly.
- **Approximate steady state.** See §3: acceptance is a tightened, enforced steady-state gate (not an exact converged state).
- **Pressure-tap sensitivity.**  $R_h$  depends on the tap planes (§4). On the corrected Large field, the default planes give  $R_h = 0.987$ ; moving them just outside the wafer ends ( $y = 0.1175/0.2525$ ) gives 1.070 (+8.4%) and farther out ( $y = 0.0775/0.2975$ ) gives 0.945 (−4.2%). Only the ordering is tap-robust.
- **Grid/timestep convergence (Small, demonstrated).** A three-mesh refinement and a timestep halving were run for the Small case (§6): timestep error is negligible ( $< 0.02\%$ ) and  $R_h$  is monotone-converging with  $\sim 0.6\%$  residual at the production mesh, so absolute  $R_h$  is good to  $\sim 1\%$  for Small. Medium and Large have not yet had the same refinement, so their absolute values carry at least a comparable (unquantified) grid uncertainty; the large separation between the three  $R_h$  makes the ordering robust regardless.
- **Reproducibility, not independent verification.** The rerun harness imports the production `Solver`; it checks reproducibility and cannot expose an error shared by that implementation.

## 11 Conclusions

The MAC projection solver is discretely consistent and mass-conserving to roundoff. The equal- $Q$  root solve reliably drives  $Q_{\text{rack}}$  to the common target, and the corrected code gives hydraulic resistances  $R_h = 14.740 > 2.728 > 0.987$  (Small  $>$  Medium  $>$  Large) *for the present discretization*, essentially unchanged by the inlet correction. With the inlet slots grid-aligned, the corrected inlet speeds are  $U_{\text{in}} \approx 0.694/0.352/0.206$  (Small unchanged; Medium and Large lower than the pre-alignment 0.388/0.237, because the earlier slots were 10–13% undersized). These remain reproduced computations, not yet mesh-independent, physically validated quantities (§10); a grid-refinement and timestep study is the next step before the absolute  $U_{\text{in}}$ ,  $\Delta p$  and  $R_h$  can be called discretization-independent. The absolute speeds also differ from the legacy finite-difference model’s (0.5605/0.3000/0.1833) because the numeric target  $Q_0$  was carried over as a fixed constant rather than redefined self-consistently for the MAC discretization.